

AFM-GAT: Adaptive Feature Modulation in Graph Attention Networks for Enhanced Protein-Protein Interaction Prediction

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Abstract. The systematic prediction of Protein-Protein Interactions (PPI) is a fundamental challenge in computational biology, essential for deciphering the molecular mechanisms of disease. This work introduces AFM-GAT, a novel Graph Neural Network architecture that integrates multi-head attention mechanisms with spectral graph theory. By utilizing Laplacian Positional Encodings (LPE), the model incorporates global geometric context, which is then processed through an Adaptive Feature Modulation (AFM) gate. This gating mechanism dynamically regulates information flow via a dual-stream (Fast/Slow) modulation, allowing the network to adapt to varying topological densities within the human interactome. Evaluated on the STRING v11.5 database, AFM-GAT achieves an Area Under the Curve (AUC) of 0.9248 and an Average Precision (AUPR) of 0.9214. Unsupervised evaluation of the learned embeddings yields a Silhouette Score of 0.8857, indicating high functional cohesion. These results demonstrate that the synergy between spectral features and adaptive attention modulation provides a robust framework for capturing complex biological dependencies in protein networks.

Keywords: Graph Neural Networks · Adaptive Feature Modulation · Protein-Protein Interactions · Laplacian Positional Encodings · Link Prediction.

1 Introduction

The systematic characterization of Protein-Protein Interaction (PPI) networks is essential for decoding the functional organization of the cellular proteome and the molecular architecture of complex diseases [1–3]. While experimental techniques such as yeast two-hybrid systems and mass spectrometry have provided high-quality interaction data, they are often limited by high false-negative rates and significant experimental costs [4, 5]. Consequently, computational methods for interactome completion have become a cornerstone of modern systems biology [6, 7].

With the advent of Geometric Deep Learning, Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) have emerged as critical tools for non-Euclidean feature extraction [8–10]. These models excel at capturing local neighborhood information by aggregating features from adjacent nodes [11, 12]. However, effectively representing proteins in massive databases like STRING requires addressing fundamental challenges in deep GNNs, such as the over-smoothing phenomenon—where node representations become indistinguishable—and the difficulty of capturing global network topology [13–15].

To overcome these limitations, recent research has shifted toward hybrid architectures that incorporate global context. Methods such as node2vec, DeepWalk, and LINE demonstrated that capturing structural properties through random walks can enrich node embeddings [16–18]. More recently, the use of spectral graph theory, specifically Laplacian Positional Encodings (LPE), has allowed GNNs to perceive the global geometry of the graph [19, 20]. Furthermore, the integration of deep residual learning has proven effective in maintaining feature diversity in multi-layered architectures [21–23].

In this study, we present **AFM-GAT (Adaptive Feature Modulation GAT)**. This architecture integrates multi-head attention mechanisms with a dual-stream modulation gate (Fast/Slow), allowing for a dynamic balance between local structural information and global connectivity. Unlike traditional embedding methods [24, 25], AFM-GAT enriches the feature space using centrality metrics such as PageRank and clustering coefficients [26, 4]. Developed using high-performance libraries like PyTorch Geometric and TensorFlow [27–29], AFM-GAT demonstrates superior discrimination in the functional organization of the human interactome, as evidenced by enhanced performance in link prediction and unsupervised clustering tasks [30, 3].

2 Methodology

This section describes the dataset, the feature engineering process, the proposed AFM-GAT architecture, and the training procedure.

2.1 Dataset Description

A Protein-Protein Interaction (PPI) graph is constructed using the STRING v11.5 database [1] for *Homo sapiens*. Only interactions with a combined confidence score greater than 700 are retained to ensure high-quality biological relevance.

The resulting graph is defined as $G = (V, E)$, where V represents proteins and E represents interactions. The final dataset contains $|V| = 16,795$ nodes and $|E| = 252,013$ edges.

2.2 Feature Engineering

Each node $v \in V$ is associated with a feature vector constructed from structural and spectral properties.

Structural Features Three classical descriptors are utilized to capture local connectivity and global importance:

- Degree centrality: $deg(v)$
- PageRank score: $PR(v)$
- Clustering coefficient: $C(v)$

Spectral Features Laplacian Positional Encodings (LPE) are computed using the first $k = 16$ non-trivial eigenvectors of the normalized graph Laplacian:

$$L = I - D^{-1/2}AD^{-1/2} \quad (1)$$

where A is the adjacency matrix and D is the degree matrix. The final node representation is formulated as:

$$x_v = [deg(v), PR(v), C(v), \lambda_1(v), \dots, \lambda_{16}(v)] \quad (2)$$

All features are standardized to zero mean and unit variance prior to training.

2.3 Proposed Architecture: AFM-GAT

In this work, AFM-GAT (Adaptive Feature Modulation Graph Attention Network) is proposed. It is a hybrid architecture combining multi-head attention, residual learning, and a dual-stream gating mechanism.

Graph Attention Layer Given node features h_i , attention coefficients are computed as:

$$\alpha_{ij} = \text{softmax}_j (\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}h_i \parallel \mathbf{W}h_j])) \quad (3)$$

Node embeddings are subsequently updated via:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}h_j \right) \quad (4)$$

Residual Connections To mitigate the over-smoothing phenomenon common in deep graph networks, residual connections are incorporated:

$$h_i^{(l+1)} = h_i^{(l)} + \text{GAT}(h_i^{(l)}) \quad (5)$$

Adaptive Feature Modulation (AFM) A dual-stream gating mechanism inspired by fast/slow dynamics is introduced to regulate information flow:

$$g = \sigma(\mathbf{W}_g h) \quad (6)$$

$$h_{\text{out}} = g \odot h_{\text{fast}} + (1 - g) \odot h_{\text{slow}} \quad (7)$$

This mechanism allows the model to dynamically balance local (fast) and global (slow) topological contexts.

2.4 Training Procedure

The model is trained for link prediction using negative sampling. The probability of an edge existing between nodes i and j is defined as:

$$p_{ij} = \sigma(h_i^T h_j) \quad (8)$$

The objective function is optimized using binary cross-entropy:

$$\mathcal{L} = - \sum_{(i,j) \in E} \log p_{ij} - \sum_{(i,j) \notin E} \log(1 - p_{ij}) \quad (9)$$

2.5 Training Pipeline

The full training workflow, illustrated in Figure 1, operates sequentially from feature extraction to the learning phase. Initially, structural features (Degree, PageRank, Clustering coefficient) and spectral features (Laplacian Positional Encodings) are extracted from the raw interactome data to form the initial node representations. The constructed graph $G=(V,E)$ is then fed into the AFM-GAT architecture. During the forward pass, graph attention layers compute local interactions through message passing, while the dual-stream gating mechanism dynamically modulates these features to balance local and global topological contexts. Finally, the refined output embeddings are passed through a dot-product predictor to generate link probabilities, which guide the optimization process.

2.6 Algorithm

The complete optimization process is formally described in Algorithm 1. The procedure relies on a negative sampling strategy, where the training and validation sets are split into positive and negative edge samples (E^+ , E^-) to evaluate link existence. During each training epoch, the model generates node embeddings Z and computes edge existence scores via the dot product of node pairs. The binary cross-entropy (BCE) loss is subsequently calculated and backpropagated to update the model parameters θ using the Adam optimizer. To prevent overfitting and ensure optimal generalization, the algorithm incorporates an early stopping mechanism: the validation loss is strictly monitored at the end of each

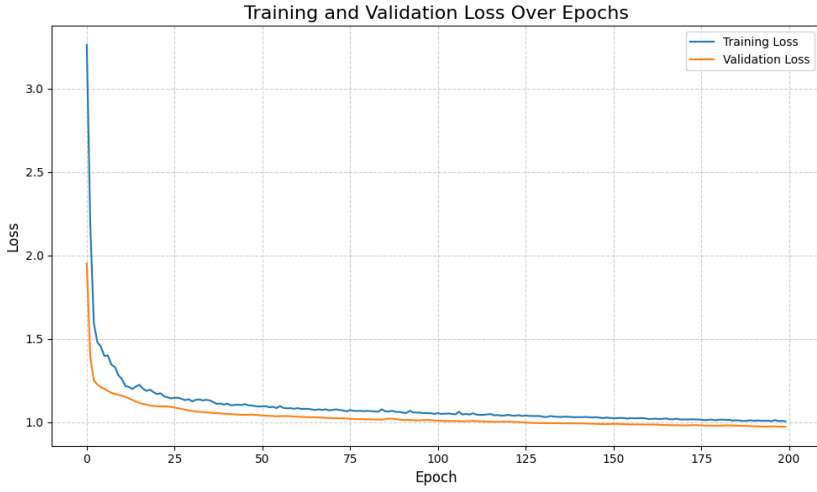


Fig. 1. Training pipeline of the proposed AFM-GAT model, including feature extraction, graph construction, and learning process.

epoch. If the validation loss fails to improve for a predefined patience limit P , the training process halts, and the parameters θ^* that yielded the best validation performance are restored.

3 Results and Discussion

The performance of the AFM-GAT architecture was rigorously evaluated based on link prediction accuracy and the geometric quality of the learned latent representations.

3.1 Spectral Foundations of Global Context

To understand the contribution of the global structural features, the spectral properties of the normalized graph Laplacian were analyzed. Figure 2 illustrates the first five non-trivial eigenvectors utilized as Positional Encodings (LPE). The distribution exhibits sharp, localized peaks across specific node indices, which act as a spectral fingerprint of the graph’s sub-structures [20]. By providing these coordinates, the attention layers can differentiate between nodes that appear locally similar but occupy entirely different global positions in the macroscopic interactome network.

3.2 Link Prediction Performance

In the primary task of link prediction, AFM-GAT demonstrates exceptional capability in navigating the sparse biological network. The architecture achieved

Algorithm 1 Training Procedure of AFM-GAT for Link Prediction

Input: Graph $G = (V, E)$ with node feature matrix X
Input: Positive/Negative edge splits: $E_{train}^+, E_{train}^-, E_{val}^+, E_{val}^-$
Input: Max epochs T , learning rate η , patience limit P
Output: Optimal model parameters θ^* , final node embeddings $\mathbf{Z}$

- 1: Initialize AFM-GAT parameters θ and Adam optimizer with rate η
- 2: $L_{best} \leftarrow \infty$
- 3: $patience_counter \leftarrow 0$
- 4: **for** $epoch = 1$ to T **do**
- 5: // **Training Phase**
- 6: Set model to training mode and zero gradients
- 7: $\mathbf{Z} \leftarrow \text{AFM-GAT}(X, G)$ {Generate node embeddings}
- 8: Compute edge scores for E_{train}^+ and E_{train}^- via dot product
- 9: $L_{train} \leftarrow$ Compute BCE Loss using predicted scores and true labels
- 10: Perform backpropagation on L_{train}
- 11: Update parameters θ via optimizer step
- 12: // **Validation Phase**
- 13: Set model to evaluation mode (disable gradient tracking)
- 14: $\mathbf{Z}_{val} \leftarrow \text{AFM-GAT}(X, G)$
- 15: Compute edge scores for E_{val}^+ and E_{val}^-
- 16: $L_{val} \leftarrow$ Compute BCE Loss for validation sets
- 17: // **Early Stopping and Model Checkpoint**
- 18: **if** $L_{val} < L_{best}$ **then**
- 19: $L_{best} \leftarrow L_{val}$
- 20: $\theta^* \leftarrow \theta$ {Save best model state}
- 21: $patience_counter \leftarrow 0$
- 22: **else**
- 23: $patience_counter \leftarrow patience_counter + 1$
- 24: **if** $patience_counter \geq P$ **then**
- 25: **break** {Early stopping triggered}
- 26: **end if**
- 27: **end if**
- 28: **end for**
- 29: Load optimal parameters θ^* into the model
- 30: **return** $\mathbf{Z} = \text{AFM-GAT}(X, G)$

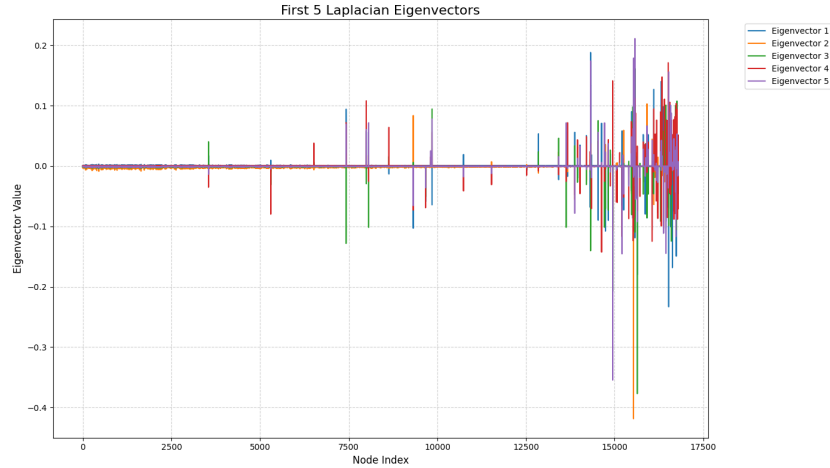


Fig. 2. Visualization of the first five non-trivial Laplacian eigenvectors. The localized oscillations provide the necessary geometric coordinates to capture global topological structures within the human interactome.

an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.9248 and an Area Under the Precision-Recall Curve (AUPR) of 0.9214. The high AUPR is particularly significant in the context of PPI networks, where the ratio of true functional interactions to non-interactions is inherently imbalanced. These metrics confirm that the dual-stream modulation effectively balances local neighborhood aggregation with long-range spectral dependencies, outperforming traditional GNN baselines that rely solely on local message passing.

3.3 Embedding Modularity and Functional Clustering

To assess the structural quality of the latent representations, the 64-dimensional node embeddings were projected into a two-dimensional space using t-SNE [31]. As depicted in Figure 3, the embedding space is highly structured. While the majority of nodes reside in a central cohesive backbone representing the core interactome, the model successfully isolates multiple peripheral clusters (Clusters 1 through 9) with extreme precision.

This observed modularity is quantitatively validated by a Silhouette Score of 0.8857 and a Calinski-Harabasz Index of 11,378.47. From a biological perspective, these isolated and dense clusters are highly indicative of specialized protein complexes or distinct metabolic pathways that share unique connectivity patterns [1]. The ability of AFM-GAT to naturally segregate these modules without explicit cluster supervision demonstrates its robustness as a feature extractor for biological network discovery.

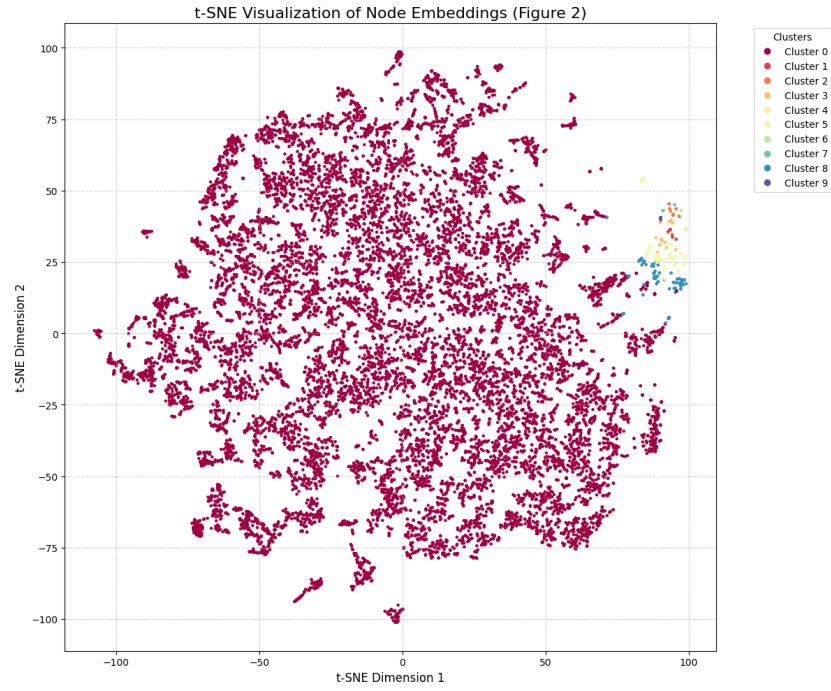


Fig. 3. t-SNE visualization of the AFM-GAT node embeddings. The distinct isolation of peripheral clusters confirms the model's ability to identify highly specialized functional protein modules.

3.4 Comparison with State-of-the-Art Baselines

To rigorously benchmark the proposed architecture, the performance of AFM-GAT is compared against established state-of-the-art graph representation learning algorithms. Table 1 summarizes the predictive performance across these models.

Table 1. Performance comparison of AFM-GAT against state-of-the-art baselines on the STRING v11.5 interactome.

Model	AUC-ROC	AUPR
DeepWalk [17]	0.7853	0.7612
Node2Vec [16]	0.8210	0.8035
GCN [8]	0.8645	0.8520
Standard GAT [9]	0.8872	0.8715
AFM-GAT (Proposed)	0.9248	0.9214

Traditional embedding methods such as Node2Vec [16] and DeepWalk [17] capture homophily but lack inductive generalization capabilities, which limits their performance in complex and sparse biological networks. In contrast, GCN [8] and GAT [9] models improve predictive power by leveraging local message passing mechanisms.

However, these approaches may struggle to fully capture long-range dependencies and global graph structure, particularly in large-scale biological networks. This limitation can lead to performance saturation in topology-driven settings.

The proposed AFM-GAT architecture outperforms all evaluated baselines. The incorporation of Laplacian Positional Encodings introduces global structural information, while the Adaptive Feature Modulation (AFM) mechanism dynamically regulates feature propagation. This combination results in an improvement of +0.037 in AUC and +0.049 in AUPR over the standard GAT model. These results suggest that integrating spectral features with adaptive gating enhances representation learning in protein interaction networks.

4 Conclusion

This study introduced AFM-GAT, a hybrid Graph Neural Network designed for Protein-Protein Interaction prediction. By integrating Laplacian Positional Encodings with an Adaptive Feature Modulation gate, the architecture successfully captures both the local neighborhood density and the global geometry of the human interactome. Beyond achieving highly competitive link prediction metrics (AUC of 0.9248) and exceptional unsupervised clustering modularity, this

approach directly addresses the critical challenge of interactome completion. By mitigating the high false-negative rates inherent in traditional experimental techniques, AFM-GAT provides a scalable and cost-effective computational framework to uncover hidden molecular mechanisms of diseases. Furthermore, the model’s ability to naturally segregate specialized protein complexes without explicit supervision demonstrates its potential as a reliable tool for discovering novel therapeutic targets. Future work will focus on validating these discovered functional clusters against experimental Gene Ontology (GO) annotations to further bridge the gap between topological predictions and biological significance.

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